



Conditional Statements Problem Bank (Java)

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Loops (Control Statements).

Section 1: Basic Decision Making

★ Positive, Negative, or Zero

★ Even or Odd

★ Maximum of Two Numbers

★ Maximum of Three Numbers

★ Smallest of Three Numbers

Absolute Value

★ Eligibility to Vote

★ Age Group Classifier

Section 2: Character-Based Problems

★ Character Type Checker using Character Class

★ Vowel or Consonant

★ Uppercase or Lowercase

Character Case Converter

★ Character Type Classifier

★ Character Type using ASCII Values

Character Category Counter

Section 3: Divisibility and Number Validation

Divisible by 5

★ Divisible by 5 and 11

Multiple of 3 or 7

Three-Digit Number Validator

Four-Digit Number Validator

★ Positive Even Number Checker

★ Leap Year Checker

Leap Century Year Checker

★ Number Range Validator

★ Divisibility Classification



Section 4: Coordinate Geometry Problems

★ Quadrant Identifier

Point on X-Axis or Y-Axis
Origin Checker

★ Axis or Quadrant Identifier

Distance-Based Zone Classification
Delivery Charge Calculator
Safe Zone Checker

Section 5: Academic and Student Systems

★ Grade Calculator

★ Pass or Fail

★ Student Result Classification

★ Scholarship Eligibility

★ Attendance Eligibility

Internal Mark Validation
Student Performance Category

★ Subject Pass Validation

Best Subject Finder

★ Exam Eligibility Checker

Section 6: Triangle Problems

★ Triangle Validity Checker

★ Triangle Type Identifier

★ Right Angled Triangle Checker

Equilateral Triangle Checker
Isosceles Triangle Checker
Scalene Triangle Checker
Largest Angle Type of Triangle

★ Angle Sum Validation

Triangle Perimeter Category

Section 7: Financial Calculations

★ Profit or Loss Calculator

★ Discount Calculator

★ Salary Bonus Calculator



★ **Electricity Bill Calculator**

★ **Income Tax Calculator**

Road Tax Calculator

GST Calculator

★ **Loan Eligibility Checker**

★ **EMI Eligibility Checker**

Mobile Recharge Offer Eligibility

Section 8: Health and Lifestyle Applications

★ **BMI Calculator**

Temperature Category

★ **Fitness Eligibility Checker**

Daily Water Intake Recommendation

Calorie Requirement Category

Sleep Quality Analyzer

Heart Rate Category

★ **Health Risk Assessment**

Step Goal Achievement

Diet Plan Recommendation

Section 9: Banking and Transaction Systems

★ **ATM Withdrawal Validator**

Account Balance Checker

★ **Minimum Balance Validator**

Loan Eligibility Checker (Bank Rules)

★ **Credit Card Eligibility**

EMI Eligibility Checker (Bank Rules)

★ **UPI Transaction Limit Check**

Bank Account Type Classifier

Transaction Fee Calculator

★ **Online Payment Validation**

Account Status Checker

Daily Withdrawal Limit Check

Section 10: Switch Statement & Switch Expression Problems (Java-Specific)

★ **Day of Week Finder using switch**

★ **Month Days Checker using switch**



- ★ **Grade Calculator using Switch Expression (yield)**
- ★ **Menu-Driven Calculator using switch**
 - Traffic Light Signal Simulator using switch
- ★ **Season Identifier using Enhanced Switch (Java 14+)**
 - Vehicle Type Fare Selector using switch
 - Multi-Case Label Grouping in switch

Section 11: Utility Applications

- ★ **Calculator using Conditional Statements**
 - Shape Area Calculator
- ★ **Movie Ticket Pricing System**
- ★ **Password Validator**
- ★ **User Access Control**
- ★ **Parking Fee Calculator**
 - Delivery Charge Calculator (General)
 - Service Subscription Checker
 - Customer Support Priority
- ★ **Service Eligibility Checker**

Section 12: Advanced Logical Problems

- ★ **Greatest of Four Numbers**
 - Marks Comparison using Ternary Operator
- ★ **Roots of Quadratic Equation**
- ★ **Discount Eligibility Checker**
- ★ **Loan Approval System**
- ★ **Multi-Level Access System**
- ★ **Insurance Eligibility Checker**
- ★ **Employee Promotion Eligibility**
 - Smart Traffic Fine Calculator
 - Membership Upgrade Checker
- ★ **Job Application Shortlisting**
 - Cab Fare Category